



# Maintenance Schedule

Diesel Engine

EPA/CARB

12V/16V4000S83

12V4000S83L

Application Group 4D

MSE50205/01E



A Rolls-Royce  
solution

| Engine model | kW/cyl.       | Application group                                       |
|--------------|---------------|---------------------------------------------------------|
| 12V4000S83   | 139,8 kW/cyl. | 4D, Frac-operation (continuous operation with low load) |
| 12V4000S83L  | 155,4 kW/cyl. | 4D, Frac-operation (continuous operation with low load) |
| 16V4000S83   | 139,8 kW/cyl. | 4D, Frac-operation (continuous operation with low load) |

*Table 1: Applicability*

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# 1 Maintenance Schedule

## 1.1 Preface

This manual provides instructions for your product from the manufacturer MTU.

### Definition of MTU

MTU refers to Rolls-Royce Power Systems AG and MTU Friedrichshafen GmbH or an affiliated company pursuant to Section § 15 AktG (German Stock Corporation Act) or a controlled company (joint venture).

### Maintenance concept

The maintenance schedule is applicable to products certified in accordance with U.S. emissions regulations and operated in the area of applicability of these regulations.

Information about the latest maintenance schedule is available at: <http://www.mtu-solutions.com>.

The following instructions must be observed for these products:

The maintenance schedule is an integral part of the emissions certification. Nonobservance of the maintenance instructions can result in violation of the statutory emissions regulations. Modification or removal of mechanical or electronic components or the installation of additional components as well as the execution of calibration processes that might affect the emissions characteristics of the product are prohibited by emissions regulations.

Although the warranty obligations are not dependent upon the use of any particular brand of replacement parts, it is recommended that emission-relevant components be serviced, replaced or repaired using only MTU-approved components or their equivalent. Emission-relevant components include, but are not limited to, control units, data records, sensors, injectors, exhaust turbo chargers, exhaust flaps and all components of the exhaust aftertreatment systems. Use of replacement parts that are not of equivalent quality to MTU-approved components may impair the effectiveness of emission control systems. The maintenance, replacement, or repair of the emission control devices and systems may be performed by any repair establishment or individual of the owner's choosing.

The maintenance system for MTU products is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of equipment availability. The maintenance intervals as well as the required scopes of maintenance tasks are based on operational experience.

Adherence to the specified maintenance intervals is essential to maintain product safety. Additional maintenance work and/or changes to the maintenance intervals may be required in the case of difficult operational and ambient conditions.

The intervals according to which the maintenance tasks have to be carried out are specified as operating hours and time limits. Whichever value is reached first shall apply.

The maintenance intervals are determined and verified by the load profile. Load profiles can be evaluated by the Service Partner of MTU. Reading out load profile data is recommended for the first time after between 500 and 1000 operating hours depending on the application concerned, or after changing the duty profile or operating area.

The individual maintenance intervals are assigned to the qualification levels QL1 to QL4.

- QL1: Operational monitoring and maintenance work which does not require disassembly of the product.
- QL2: Exchange of components and parts in case of repair (corrective only, not part of the maintenance schedule).
- QL3: Maintenance work which requires partial disassembly of the product.
- QL4: Maintenance work which requires complete disassembly of the product.

## Maintenance task tolerances

Maintenance tasks shall be completed within certain tolerances comprising a percentage and a maximum value, whereby the smaller value shall apply. Taking full advantage of such tolerances does not influence the scheduled timing of the ensuing maintenance task.

| Interval (hrs)/limit | Limit in %           | Limit max.                                                           |
|----------------------|----------------------|----------------------------------------------------------------------|
| Operating hours      | ± 10%                | ± 500 hrs                                                            |
| Years                | ± 10%                | ± 30 days                                                            |
| Example 1            | Task every 1000 hrs  | Between 900 hrs and 1100 hrs (10% of 1000 hrs)                       |
| Example 2            | Task every 30000 hrs | Between 29500 hrs and 30500 hrs (10% of 30000 hrs, but max. 500 hrs) |

## Additional notes on maintenance and preservation:

The relevant component manufacturer's instructions apply with respect to the maintenance of any components which do not appear in this maintenance schedule.

For the change intervals of fluids and lubricants, refer to the corresponding Fluids and Lubricants Specifications.

If the engine is to remain out of service for a longer time period, MTU recommends to carry out a monthly test run until engine operating temperature is reached. If the engine is scheduled to remain out of service for > 1 month, carry out engine preservation procedures in accordance with the Preservation and Represervation Specifications

The latest version of the specifications is available at <http://www.mtu-solutions.com>.

## 1.2 Allocation matrix application group 4D

| V 4000 S83 / S83 L       |        |               |                  |                 |                   |   |   |
|--------------------------|--------|---------------|------------------|-----------------|-------------------|---|---|
| Load profile             |        | Load factor % | Load indicator % | 0 - 140 kW/cyl. | 141 - 156 kW/cyl. | - | - |
| Load %                   | Time % |               |                  |                 |                   |   |   |
| 110 <sub>(100-110)</sub> | —      | 0 - 41        | ≤ 9              | 13,500          | 10,000            | — | — |
| 100 <sub>(90-100)</sub>  | 5      |               |                  |                 |                   |   |   |
| 90 <sub>(80-90)</sub>    | —      |               |                  |                 |                   |   |   |
| 80 <sub>(65-80)</sub>    | 25     |               |                  |                 |                   |   |   |
| 65 <sub>(50-65)</sub>    | 20     |               |                  |                 |                   |   |   |
| 50 <sub>(30-50)</sub>    | —      |               |                  |                 |                   |   |   |
| 30 <sub>(5-30)</sub>     | —      |               |                  |                 |                   |   |   |
| Idle <sub>(0-5)</sub>    | 50     |               |                  |                 |                   |   |   |

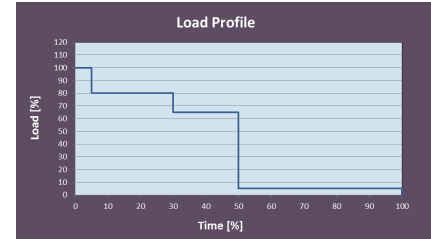


Table 2: Allocation matrix application group 4D

## 1.3 Maintenance schedule matrix 13,500 h

### 0 to 13,500 Operating hours

| Task   | Limit | Operating hours [h] |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        |   |   |
|--------|-------|---------------------|-----|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--------|--------|--------|--------|--------|--------|--------|--------|---|---|
|        |       | Daily               | 500 | 1,000 | 1,500 | 2,000 | 2,500 | 3,000 | 3,500 | 4,000 | 4,500 | 5,000 | 5,500 | 6,000 | 6,500 | 7,000 | 7,500 | 8,000 | 8,500 | 9,000 | 9,500 | 10,000 | 10,500 | 11,000 | 11,500 | 12,000 | 12,500 | 13,000 | 13,500 |   |   |
| Engine |       |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        |   |   |
| W0800  | -     |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        |   |   |
| W1024  | 1 w   |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        |   |   |
| W1008  | 2 a   |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        |   |   |
| W0500  | -     | X                   |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        |   |   |
| W0501  | -     | X                   |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        |   |   |
| W0503  | -     | X                   |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        |   |   |
| W0505  | -     | X                   |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        |   |   |
| W0506  | -     | X                   |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        |   |   |
| W0525  | -     | X                   |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        |   |   |
| W1009  | 2 a   |                     | X   | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X      | X      | X      | X      | X      | X      | X      | X      | X | X |
| W1864  | 6 m   |                     |     | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X      |        | X      |        | X      |        | X      |        |   |   |
| W1463  | 1 a   |                     |     | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X      |        | X      |        | X      |        | X      |        |   |   |
| W1001  | 2 a   |                     |     | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X      |        | X      |        | X      |        | X      |        |   |   |
| W1241  | 2 a   |                     |     | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X      |        | X      |        | X      |        | X      |        |   |   |
| W1575  | 1 a   |                     |     |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |       |        |        |        |        | X      |        |        |        |   |   |
| W1207  | 2 a   |                     |     | X     |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |       |        |        |        |        | X      |        |        |        |   |   |
| W1046  | 2 a   |                     |     |       |       |       |       |       |       |       | X     |       |       |       |       |       |       |       |       | X     |       |        |        |        |        |        |        |        | X      |   |   |
| W1011  | 6 a   |                     |     |       |       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       | X     |       |        |        |        |        |        |        |        | X      |   |   |
| W1026  | 6 a   |                     |     |       |       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       | X     |       |        |        |        |        |        |        |        | X      |   |   |
| W1006  | 9 a   |                     |     |       |       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       | X     |       |        |        |        |        |        |        |        | X      |   |   |
| W1713  | 9 a   |                     |     |       |       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       | X     |       |        |        |        |        |        |        |        | X      |   |   |
| W1689  | 9 a   |                     |     |       |       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       | X     |       |        |        |        |        |        |        |        | X      |   |   |
| W1610  | 9 a   |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        | X      |   |   |
| W1250  | 5 a   |                     |     |       |       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       | X     |       |        |        |        |        |        |        |        | X      |   |   |
| W1251  | 6 a   |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        | X      |   |   |
| W2000  | 18 a  |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        | X      |   |   |
| W2001  | 18 a  |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        | X      |   |   |
| W2002  | 18 a  |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        | X      |   |   |
| W2004  | 18 a  |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        | X      |   |   |
| W2007  | 18 a  |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        | X      |   |   |
| W2009  | 18 a  |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        | X      |   |   |
| W2010  | 18 a  |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        | X      |   |   |
| W2062  | 18 a  |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        | X      |   |   |
| W2070  | 18 a  |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        | X      |   |   |
| W2110  | 18 a  |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        | X      |   |   |
| W2154  | 18 a  |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        | X      |   |   |
| W1051  | 18 a  |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        | X      |   |   |
| W1043  | 18 a  |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        | X      |   |   |
| W1134  | 18 a  |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        | X      |   |   |
| W1605  | 18 a  |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        | X      |   |   |
| W1735  | 18 a  |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        | X      |   |   |
| W1041  | 18 a  |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        | X      |   |   |

w = weeks  
m = months  
a = years

| Task                                 | Limit | Operating hours [h] |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        |   |  |  |
|--------------------------------------|-------|---------------------|-----|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--------|--------|--------|--------|--------|--------|--------|--------|---|--|--|
|                                      |       | Daily               | 500 | 1,000 | 1,500 | 2,000 | 2,500 | 3,000 | 3,500 | 4,000 | 4,500 | 5,000 | 5,500 | 6,000 | 6,500 | 7,000 | 7,500 | 8,000 | 8,500 | 9,000 | 9,500 | 10,000 | 10,500 | 11,000 | 11,500 | 12,000 | 12,500 | 13,000 | 13,500 |   |  |  |
| W1035                                | 6 a   |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        | X |  |  |
| W3000                                | 18 a  |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        | X |  |  |
| W3001                                | 18 a  |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        | X |  |  |
| W3002                                | 18 a  |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        | X |  |  |
| W3003                                | 18 a  |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        | X |  |  |
| W3004                                | 18 a  |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        | X |  |  |
| W3005                                | 18 a  |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        | X |  |  |
| W3007                                | 18 a  |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        | X |  |  |
| W3018                                | 18 a  |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        | X |  |  |
| W3021                                | 18 a  |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        | X |  |  |
| W3083                                | 18 a  |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        | X |  |  |
| W3097                                | 18 a  |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        | X |  |  |
| W3100                                | 18 a  |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        | X |  |  |
| W3103                                | 18 a  |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        | X |  |  |
| W3108                                | 18 a  |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        | X |  |  |
| W3112                                | 18 a  |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        | X |  |  |
| W3143                                | 18 a  |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        | X |  |  |
| W3177                                | 18 a  |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        | X |  |  |
| W3182                                | 18 a  |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        | X |  |  |
| W1124                                | 18 a  |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        | X |  |  |
| W1331                                | 18 a  |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        | X |  |  |
| X5374                                | 18 a  |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        | X |  |  |
| w = weeks<br>m = months<br>a = years |       |                     |     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |        |        |        |        |        |        |        |        |   |  |  |



## 1.4 Maintenance tasks 13,500 h

| Qualification level                  | Interval [h] | Limit | Item                         | Maintenance tasks                                                                                                                                                               | Option | Task  |
|--------------------------------------|--------------|-------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------|
| Engine                               |              |       |                              |                                                                                                                                                                                 |        |       |
| QL1                                  | -            | -     | EPA label                    | Check if emission label is fitted and legible, and if it contains the correct information.                                                                                      |        | W0800 |
| QL1                                  | -            | 1 w   | Emergency air shut-off flaps | Check operation of emergency air shut-off flaps (electrical).                                                                                                                   |        | W1024 |
| QL1                                  | -            | 2 a   | Engine oil filter            | Fit new engine oil filters each time the engine oil is changed or, at the latest, on expiry of the time limit (given in years).                                                 |        | W1008 |
| QL1                                  | Daily        | -     | Engine operation             | Check engine oil level.                                                                                                                                                         |        | W0500 |
|                                      |              |       |                              | Carry out visual inspection of engine for general condition and leaks.                                                                                                          |        | W0501 |
|                                      |              |       |                              | Inspect service indicator of air filter.                                                                                                                                        |        | W0503 |
|                                      |              |       |                              | Check relief bores of coolant pump(s).                                                                                                                                          |        | W0505 |
|                                      |              |       |                              | Check for abnormal running noises, exhaust gas color and vibration.                                                                                                             |        | W0506 |
|                                      |              |       |                              | Inspect battery-charging generator for contamination and clean if necessary.                                                                                                    |        | W0525 |
| QL1                                  | 500          | 2 a   | Centrifugal oil filter       | Check thickness of oil residue layer. Clean. Fit new sleeve, at the latest, each time the engine oil is changed.                                                                | X      | W1009 |
| QL1                                  | 1000         | 6 m   | Automatic engine oil filter  | Check and clean oil indicator filter.                                                                                                                                           | X      | W1864 |
| QL1                                  | 1000         | 1 a   | Engine mounts                | Carry out visual inspection of engine mounts for general condition.                                                                                                             | X      | W1463 |
| QL1                                  | 1000         | 2 a   | Fuel filter                  | Fit new fuel filter or new fuel filter insert.                                                                                                                                  |        | W1001 |
| QL1                                  | 1000         | 2 a   | Belt drive                   | Inspect condition of drive belts and fit new ones if necessary. Adjust tension.                                                                                                 |        | W1241 |
| QL1                                  | 3000         | 1 a   | Engine mounts                | Carry out visual inspection of Trunnion mounts for general condition.                                                                                                           |        | W1575 |
| QL1                                  | 3000         | 2 a   | Valve gear                   | Check valve clearance, adjust if required. ATTENTION! Initial adjustment after 1,000 operating hours and subsequently 1,000 operating hours after each cylinder-head overhaul.. |        | W1207 |
| QL1                                  | 4500         | 2 a   | Crankcase breathers          | Fit new filters or filter inserts.                                                                                                                                              |        | W1046 |
| QL1                                  | 4500         | 6 a   | Combustion chambers          | Inspect cylinder chambers using endoscope.                                                                                                                                      |        | W1011 |
| QL1                                  | 4500         | 6 a   | Engine mounts                | Check buffer clearance of mounts, check proper seating of securing screws.                                                                                                      | X      | W1026 |
| QL1                                  | 4500         | 9 a   | Fuel injectors               | Replace fuel injectors.                                                                                                                                                         |        | W1006 |
| QL1                                  | 4500         | 9 a   | Engine governor              | Injector: reset drift compensation parameters (CDC).                                                                                                                            |        | W1713 |
| QL1                                  | 4500         | 9 a   | Cylinder heads               | Measure valve stem end to cylinder head top distance.                                                                                                                           |        | W1689 |
| QL1                                  | 13500        | 9 a   | Engine mounts                | Check securing screws of trunnion mounts for secure seating.                                                                                                                    |        | W1610 |
| QL3                                  | 4500         | 5 a   | Rubber sleeves               | Replace all rubber sleeves.                                                                                                                                                     |        | W1250 |
| QL3                                  | 13500        | 6 a   | Hose lines                   | Replace all hose lines.                                                                                                                                                         |        | W1251 |
| QL3                                  | 13500        | 18 a  | Component maintenance        | Before starting maintenance work, drain the coolant and flush the cooling systems.                                                                                              |        | W2000 |
|                                      |              |       |                              | Inspect rocker arms and valve bridge for wear. Insert an endoscope through the pushrod bore to visually inspect swing followers and camshaft running surfaces.                  |        | W2001 |
|                                      |              |       |                              | Clean air ducting.                                                                                                                                                              |        | W2002 |
|                                      |              |       |                              | Fit new high-pressure fuel sensor.                                                                                                                                              |        | W2004 |
|                                      |              |       |                              | Check charge-air coolant thermostat and fit new thermal actuator.                                                                                                               |        | W2007 |
|                                      |              |       |                              | Inspect centrifugal oil filter for wear.                                                                                                                                        | X      | W2009 |
|                                      |              |       |                              | Overhaul starter.                                                                                                                                                               |        | W2010 |
|                                      |              |       |                              | Fit new seals/sealing materials for all disassembled components.                                                                                                                |        | W2062 |
|                                      |              |       |                              | Overhaul charge-air coolant pump.                                                                                                                                               |        | W2070 |
|                                      |              |       |                              | Overhaul engine coolant pump.                                                                                                                                                   |        | W2110 |
| w = weeks<br>m = months<br>a = years |              |       |                              |                                                                                                                                                                                 |        |       |

| Qualification level | Interval [h] | Limit | Item                               | Maintenance tasks                                                                                                                       | Option | Task  |
|---------------------|--------------|-------|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|--------|-------|
|                     |              |       |                                    | Check engine coolant thermostat and check thermal actuator, fit new one if necessary.                                                   |        | W2154 |
| QL3                 | 13500        | 18 a  | Fuel delivery pump                 | Fit new fuel delivery pump.                                                                                                             |        | W1051 |
| QL3                 | 13500        | 18 a  | Battery charging alternator        | Overhaul battery-charging generator.                                                                                                    |        | W1043 |
| QL3                 | 13500        | 18 a  | Cylinder heads                     | Overhaul cylinder heads.                                                                                                                |        | W1134 |
| QL3                 | 13500        | 18 a  | Oil priming pump                   | Visually check oil priming pump.                                                                                                        |        | W1605 |
| QL3                 | 13500        | 18 a  | Auxiliary PTO                      | Replace PTO.                                                                                                                            |        | W1735 |
| QL3                 | 13500        | 18 a  | Turbochargers                      | Fit new turbochargers.                                                                                                                  |        | W1041 |
| QL4                 | 13500        | 6 a   | Automatic engine oil filter        | Remove filter housing and clean. Fit new filter candles.                                                                                | X      | W1035 |
| QL4                 | 13500        | 18 a  | Extended component maintenance     | Completely disassemble the engine. Inspect engine components as per assembly instructions and repair or fit new components as required. |        | W3000 |
|                     |              |       |                                    | Replace all elastomeric parts and seals with new ones.                                                                                  |        | W3001 |
|                     |              |       |                                    | Fit new piston rings.                                                                                                                   |        | W3002 |
|                     |              |       |                                    | Fit new conrod bearings.                                                                                                                |        | W3003 |
|                     |              |       |                                    | Fit new crankshaft bearings.                                                                                                            |        | W3004 |
|                     |              |       |                                    | Fit new cylinder liners.                                                                                                                |        | W3005 |
|                     |              |       |                                    | Fit new high-pressure fuel pump.                                                                                                        |        | W3007 |
|                     |              |       |                                    | Fit new camshaft bearings and camshaft thrust bearings.                                                                                 |        | W3018 |
|                     |              |       |                                    | Fit new pressure relief valve in high-pressure fuel system.                                                                             |        | W3021 |
|                     |              |       |                                    | Replace wiring harnesses.                                                                                                               |        | W3083 |
|                     |              |       |                                    | Fit new shaft misalignment coupling.                                                                                                    | X      | W3097 |
|                     |              |       |                                    | Fit new rubber elements of engine mounts.                                                                                               | X      | W3100 |
|                     |              |       |                                    | Replace swing followers and swing-follower shafts.                                                                                      |        | W3103 |
|                     |              |       |                                    | Check vibration damper, fit new one if necessary.                                                                                       |        | W3108 |
|                     |              |       |                                    | Replace diaphragm coupling.                                                                                                             | X      | W3112 |
|                     |              |       |                                    | Replace pulley.                                                                                                                         |        | W3143 |
|                     |              |       |                                    | Check engine oil pump, replace if necessary.                                                                                            |        | W3177 |
|                     |              |       |                                    | Gear train: Check tooth flanks for wear (visual inspection), replace bearing bushings.                                                  |        | W3182 |
| QL4                 | 13500        | 18 a  | Engine mounts                      | Fit new Trunnion mount element.                                                                                                         |        | W1124 |
| QL4                 | 13500        | 18 a  | Exhaust pipe bellows               | Replace exhaust pipe bellows.                                                                                                           |        | W1331 |
| QL4                 | 13500        | 18 a  | Steel-spring coupling (Geislinger) | Inspect steel-spring coupling, repair if necessary.                                                                                     | X      | X5374 |

w = weeks  
m = months  
a = years